
Jason Trimble

AI Specialist and Web Developer

Houston Community College — Applied AI & Robotics Program

Building practical AI systems across **computer vision**, **natural language processing**, and **machine learning** — from emotion recognition models to fraud detection analysis and NLP pipelines.

KidMood: Classroom Emotion Recognition

ITAI 1378 — Computer Vision & AI Capstone

PROBLEM & APPROACH

Teachers cannot continuously monitor every student's emotional state. KidMood uses computer vision to classify facial expressions as an assistive awareness tool.

- ▶ FER-2013 facial expression dataset
- ▶ MobileNetV2 transfer learning
- ▶ 4-class classification: Happy, Sad, Angry, Fear
- ▶ Stratified train/validation/test split
- ▶ Baseline CNN for comparison

TECHNICAL IMPLEMENTATION

TensorFlow/Keras

MobileNetV2

Transfer Learning

OpenCV

scikit-learn

NumPy

Matplotlib

4

Emotion Classes

224×224

Input Resolution

Tier 2

Transfer Learning

Full pipeline: data loading → preprocessing → training → evaluation → inference
github.com/spacemanspiff-713/KidMood-Final-Project

AI Applications & Natural Language Processing



AI in Fraud Detection

ITAI 2372 — A04 Case Study

Analyzed how **JPMorgan Chase** uses machine learning, large language models, entity extraction, and behavioral analytics to detect business email compromise and financial fraud at scale.

- LLMs
- Entity Extraction
- Behavioral Analytics
- Risk Scoring

Key finding: AI amplifies fraud professionals — it does not replace them. Strongest programs combine AI with validated data, human review, and governance.



POS Tagging & Sentiment Analysis

ITAI 2373 — Lab 05

Built an NLP pipeline using **NLTK** and **SpaCy** for part-of-speech tagging, customer service sentiment analysis, urgency detection, and tagger benchmarking across text types.

- NLTK
- SpaCy
- pandas
- matplotlib
- seaborn

Demonstrated that SpaCy provides cleaner broad-category tags for production, while NLTK excels at fine-grained Penn Treebank analysis and speed.

From Coursework to Industry Application



Fraud Detection & Financial Security

Applying ML and LLMs to detect suspicious patterns in payments, emails, and account behavior — reducing financial crime while minimizing false positives.

← [A04: JPMorgan Chase Case Study](#)



Customer Intelligence & NLP

Using POS tagging and sentiment analysis to automatically categorize support tickets, detect urgency, and route issues to the right teams.

← [L05: POS Tagging & Sentiment Lab](#)



Computer Vision for Accessibility

Building emotion-aware systems that help teachers, counselors, and caregivers notice when someone may need support — assistive AI, not replacement.

← [KidMood: Emotion Recognition](#)

Transferable skills: Data preprocessing, model evaluation, responsible AI analysis, technical communication, and end-to-end ML pipeline construction — applicable across finance, healthcare, education, retail, government, and manufacturing.

Thank You

I build AI systems that solve real problems — combining hands-on implementation with thoughtful analysis of technology's impact on people and industries.

PORTFOLIO REPOSITORY

github.com/spacemanspiff-713/Jason-Trimble-AI-Portfolio

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Microsoft AI-900: Azure AI Fundamentals — In Progress